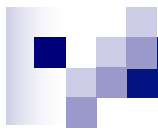


Combinatorial Mathematics



Chapter 6 Counting

- 
- 6.1 The Basics of Counting
 - 6.2 The Pigeonhole Principle
 - 6.3 Permutations and Combinations
 - 6.4 Binomial Permutations and Combinations
 - 6.5 Generating Permutations and Combination



The Basics of Counting

The Product Rule

The Sum Rule

The Subtraction Rule

The Division Rule

Examples, Examples, and Examples

Tree Diagrams



§6.1 The Basics of counting

※ A counting problem: (Example 16)

Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

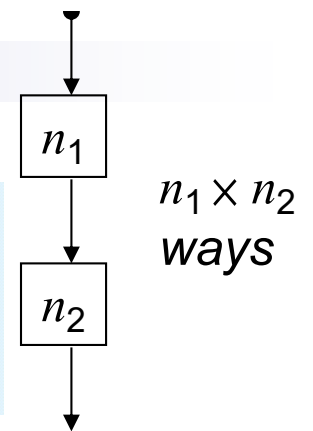
※ This section introduces

- a variety of other counting problems
- the basic techniques of counting.



✧ the Product Rule:

THE PRODUCT RULE Suppose that a procedure can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and for each of these ways of doing the first task, there are n_2 ways to do the second task, then there are $n_1 n_2$ ways to do the procedure.



EXAMPLE 1 A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solution: The procedure of assigning offices to these two employees consists of assigning an office to Sanchez, which can be done in 12 ways, then assigning an office to Patel different from the office assigned to Sanchez, which can be done in 11 ways. By the product rule, there are $12 \cdot 11 = 132$ ways to assign offices to these two employees. 

EXAMPLE 3 There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there?

Solution: The procedure of choosing a port consists of two tasks, first picking a microcomputer and then picking a port on this microcomputer. Because there are 32 ways to choose the microcomputer and 24 ways to choose the port no matter which microcomputer has been selected, the product rule shows that there are $32 \cdot 24 = 768$ ports. 

Example 2 The chair of an auditorium (大禮堂) is to be labeled with a letter and a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

Sol: $\underline{26} \times \underline{100} = 2600$ ways to label chairs.

letter

$$\begin{array}{l} 1 \leq x \leq 100 \\ x \in \mathbb{N} \end{array}$$

Example 4 How many different bit strings are there of length seven?

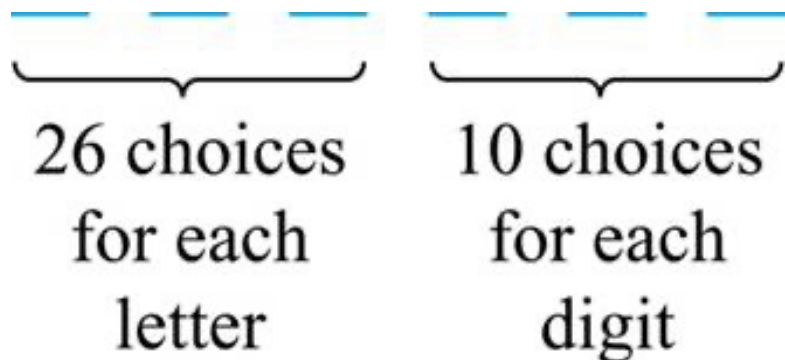
Sol:

1	2	3	4	5	6	7	
☐	☐	☐	☐	☐	☐	☐	$\rightarrow 2^7$ 種
$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	
0,1	0,1	0,1	...	...	...	0,1	

Example 5: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits?

Solution: By the product rule,

there are $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$ different possible license plates.



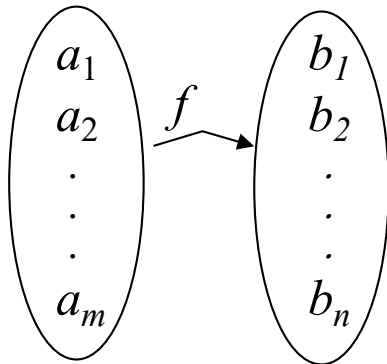
Ex.



Counting Functions

Example 6 How many functions are there from a set with m elements to one with n elements?

Sol:



$f(a_1)=?$ 可以是 $b_1 \sim b_n$, 共 n 種

$f(a_2)=?$ 可以是 $b_1 \sim b_n$, 共 n 種

$\vdots$

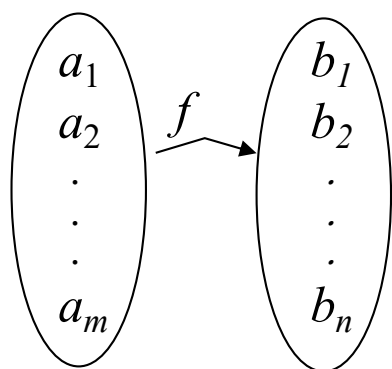
$f(a_m)=?$ 可以是 $b_1 \sim b_n$, 共 n 種

$$\therefore n^m$$

Counting Functions

Example 7 How many one-to-one functions are there from a set with m elements to one with n element?
($m \leq n$)

Sol: $f(a_1) = ?$ 可以是 $b_1 \sim b_n$, 共 n 種
 $f(a_2) = ?$ 可以是 $b_1 \sim b_n$, 但不能 $= f(a_1)$, 共 $n-1$ 種
 $f(a_3) = ?$ 可以是 $b_1 \sim b_n$, 但不能 $= f(a_1)$, 也不能 $= f(a_2)$,
共 $n-2$ 種



$\vdots$
 $f(a_m) = ?$ 不可 $= f(a_1), f(a_2), \dots, f(a_{m-1})$, 故共 $n-(m-1)$ 種
 $\therefore$ 共 $n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-m+1)$ 種 1-1 function #



Telephone Numbering Plan

Example: The *North American numbering plan (NANP)* specifies that a **telephone number** consists of 10 digits, consisting of a three-digit **area code**, a three-digit **office code**, and a four-digit **station code**. There are some restrictions on the digits.

- Let X denote a digit from 0 through 9.
- Let N denote a digit from 2 through 9.
- Let Y denote a digit that is 0 or 1.
- In the **old plan** (in use in the 1960s) the format was $NYX-NNX-XXXX$.
- In the **new plan**, the format is $NXX-NXX-XXXX$.

How many different telephone numbers are possible under the old plan and the new plan?

Solution: Use the Product Rule.

- There are $8 \cdot 2 \cdot 10 = 160$ area codes with the format NYX .
- There are $8 \cdot 10 \cdot 10 = 800$ area codes with the format NXX .
- There are $8 \cdot 8 \cdot 10 = 640$ office codes with the format NNX .
- There are $10 \cdot 10 \cdot 10 \cdot 10 = 10,000$ station codes with the format $XXXX$.

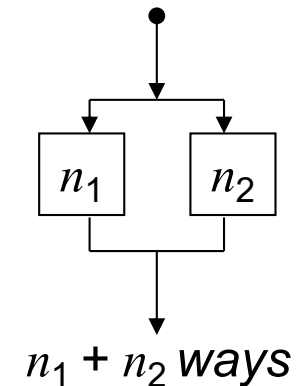
Number of **old plan** telephone numbers: $160 \cdot 640 \cdot 10,000 = 1,024,000,000$.

Number of **new plan** telephone numbers: $800 \cdot 800 \cdot 10,000 = 6,400,000,000$.

Basic counting principles

※ the Sum Rule:

If a first task can be done in n_1 ways and a second task in n_2 ways, and if these tasks cannot be done at the same time. then there are $n_1 + n_2$ ways to do either task.



Example 12

Suppose that either a member of faculty or a student is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the faculty and 83 students?

Solution: By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick a representative.



Example 13 A student can choose a computer project from one of three lists. The three lists contain 23, 15 and 19 possible projects respectively. How many possible projects are there to choose from?

Sol: $23+15+19=57$ projects.

Example. A student wishes to take either a mathematics course or a biology course, but not both. If there are four mathematics courses and three biology courses for which the student has the necessary prerequisites, then the student can choose a course to take in $4 + 3 = 7$ ways. □



Combining the Sum and Product Rule

Example 15

In a version of Basic, the name of a **variable** is a string of **one or two** alphanumeric characters, where uppercase and lowercase letters are not distinguished. Moreover, a variable name **must begin with a letter** and **must be different from the five strings of two characters** that are reserved for programming use. How many different variable names are there in this version of Basic?

Sol:

Let V_i be the number of variable names of length i .

$$V_1 = 26$$

$$V_2 = 26 \cdot 36 - 5$$

$\therefore 26 + 26 \cdot 36 - 5$ different names.



Counting Passwords

Combining the sum and product rule allows us to solve more complex problems.

Example: Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution: Let P be the total number of passwords, and let P_6 , P_7 , and P_8 be the passwords of length 6, 7, and 8.

- By the sum rule $P = P_6 + P_7 + P_8$.
- To find each of P_6 , P_7 , and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:

$$P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560.$$

$$P_7 = 36^7 - 26^7 =$$

$$78,364,164,096 - 8,031,810,176 = 70,332,353,920.$$

$$P_8 = 36^8 - 26^8 =$$

$$2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880.$$

Consequently, $P = P_6 + P_7 + P_8 = 2,684,483,063,360$.

Internet Addresses

Version 4 of the Internet Protocol (IPv4) uses 32 bits.

Bit Number	0	1	2	3	4	8	16	24	31
Class A	0	netid							hostid
Class B	1	0	netid						hostid
Class C	1	1	0	netid					hostid
Class D	1	1	1	0	Multicast Address				
Class E	1	1	1	1	0	Address			

Class A Addresses: used for the largest networks, a 0, followed by a 7-bit netid and a 24-bit hostid.

Class B Addresses: used for the medium-sized networks, a 10, followed by a 14-bit netid and a 16-bit hostid.

Class C Addresses: used for the smallest networks, a 110, followed by a 21-bit netid and an 8-bit hostid.

- Neither Class D nor Class E addresses are assigned as the address of a computer on the internet. Only Classes A, B, and C are available.
- 1111111 is not available as the netid of a Class A network.
- Hostids consisting of all 0s and all 1s are not available in any network.

[Jump to long description](#)

Counting Internet Addresses

Example: How many different IPv4 addresses are available for computers on the internet?

Solution: Use both the sum and the product rule. Let x be the number of available addresses, and let x_A , x_B , and x_C denote the number of addresses for the respective classes.

- To find, x_A : $2^7 - 1 = 127$ netids. $2^{24} - 2 = 16,777,214$ hostids.

$$x_A = 127 \cdot 16,777,214 = 2,130,706,178.$$

- To find, x_B : $2^{14} = 16,384$ netids. $2^{16} - 2 = 16,534$ hostids.

$$x_B = 16,384 \cdot 16,534 = 1,073,709,056.$$

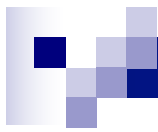
- To find, x_C : $2^{21} = 2,097,152$ netids. $2^8 - 2 = 254$ hostids.

$$x_C = 2,097,152 \cdot 254 = 532,676,608.$$

- Hence, the total number of available IPv4 addresses is

$$\begin{aligned} x &= x_A + x_B + x_C \\ &= 2,130,706,178 + 1,073,709,056 + 532,676,608 \\ &= 3,737,091,842. \end{aligned}$$

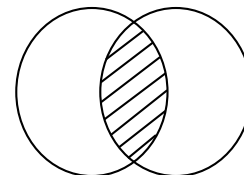
Not Enough Today !!
The newer IPv6
protocol solves the
problem of too few
addresses.



※ the Subtraction Rule: If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ **minus** the number of ways to do the **task** that are **common to the two different ways**.

Also known as, the *principle of inclusion-exclusion*:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

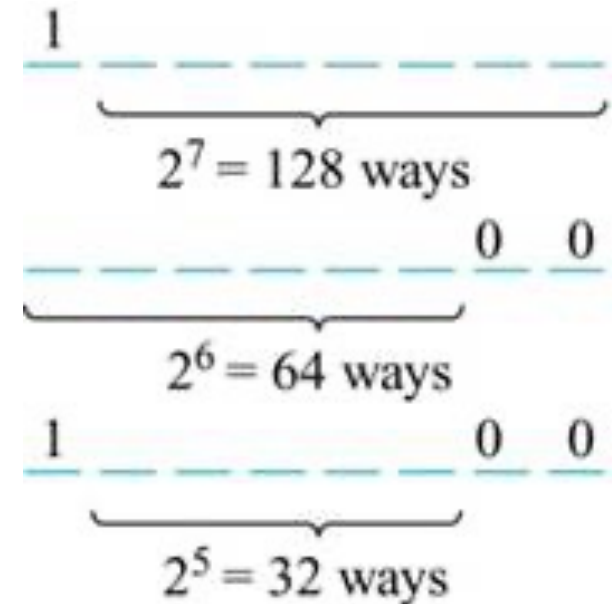


Counting Bit Strings

Example: How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution: Use the subtraction rule.

- Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
- Number of bit strings of length eight that end with bits 00: $2^6 = 64$
- Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$



Hence, the number is $128 + 64 - 32 = 160$.



✖ the Division Rule: There are “ n/d ” ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w , exactly d of the n ways correspond to way w .

Restated in terms of sets: If the finite set A is the union of n pairwise disjoint subsets each with d elements, then $n = |A|/d$.

In terms of functions: If f is a function from A to B , where both are finite sets, and for every value $y \in B$ there are exactly d values $x \in A$ such that $f(x) = y$, then $|B| = |A|/d$.

Example: How many ways are there to seat four people around a circular table, where two seatings are considered the same when each person has the same left and right neighbor?

Solution: Number the seats around the table from 1 to 4 proceeding clockwise. There are four ways to select the person for seat 1, 3 for seat 2, 2, for seat 3, and one way for seat 4. Thus, there are $4! = 24$ ways to order the four people. But since two seatings are the same when each person has the same left and right neighbor, for every choice for seat 1, we get the same seating.

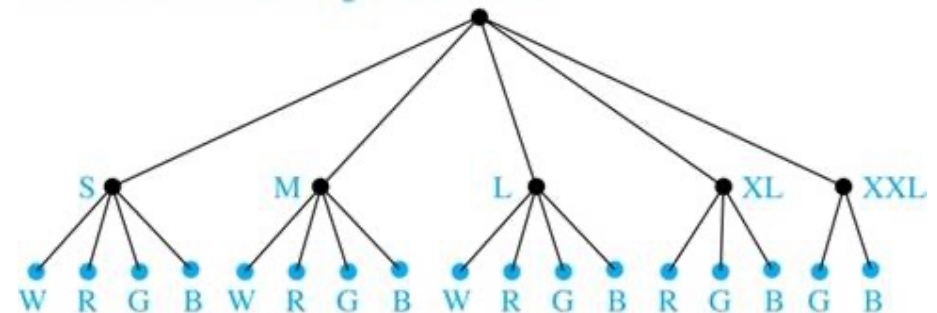
Therefore, by the division rule, there are $24/4 = 6$ different seating arrangements.

Tree Diagrams

Tree Diagrams: We can solve many counting problems through the use of tree diagrams, where a **branch** represents a *possible choice* and the **leaves** represent *possible outcomes*.

Example: Suppose that “I Love Discrete Math” T-shirts come in five different sizes: S,M,L,XL, and XXL. Each size comes in four colors (white, red, green, and black), except XL, which comes only in red, green, and black, and XXL, which comes only in green and black. What is the minimum number of shirts that the campus book store needs to stock to have one of each size and color available?

Solution: Draw the tree diagram. W = white, R = red, G = green, B = black



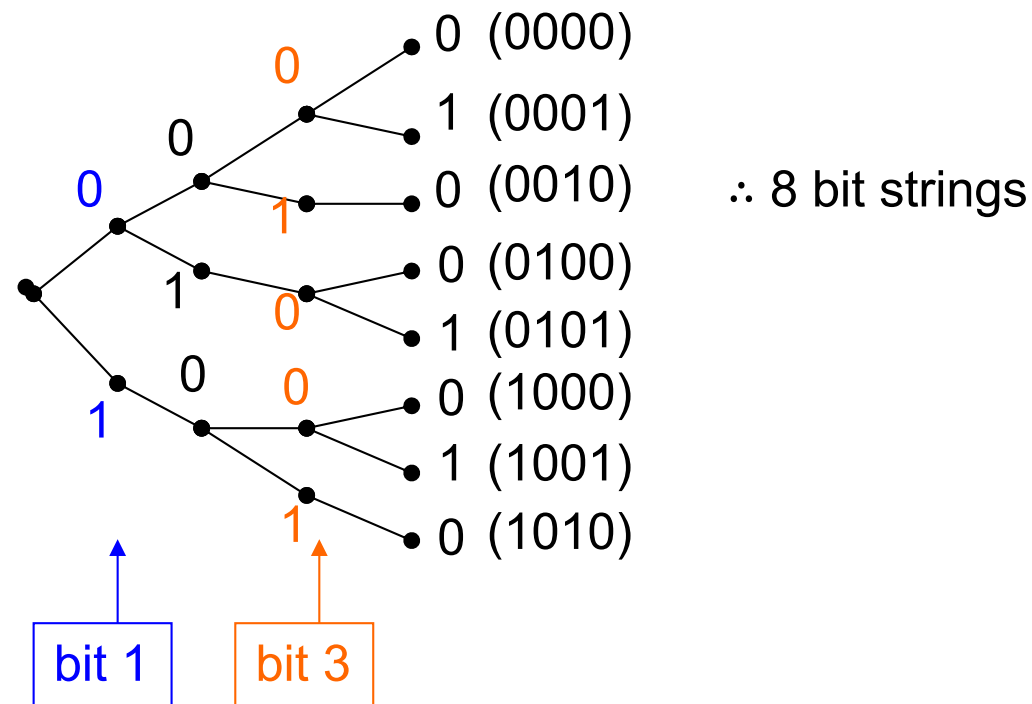
The store must stock 17 T-shirts.

[Jump to long description](#)

※ Tree Diagrams

Example 19 How many bit strings of length four *do not* have two consecutive 1s ?

Sol:



Winning team
shown in color

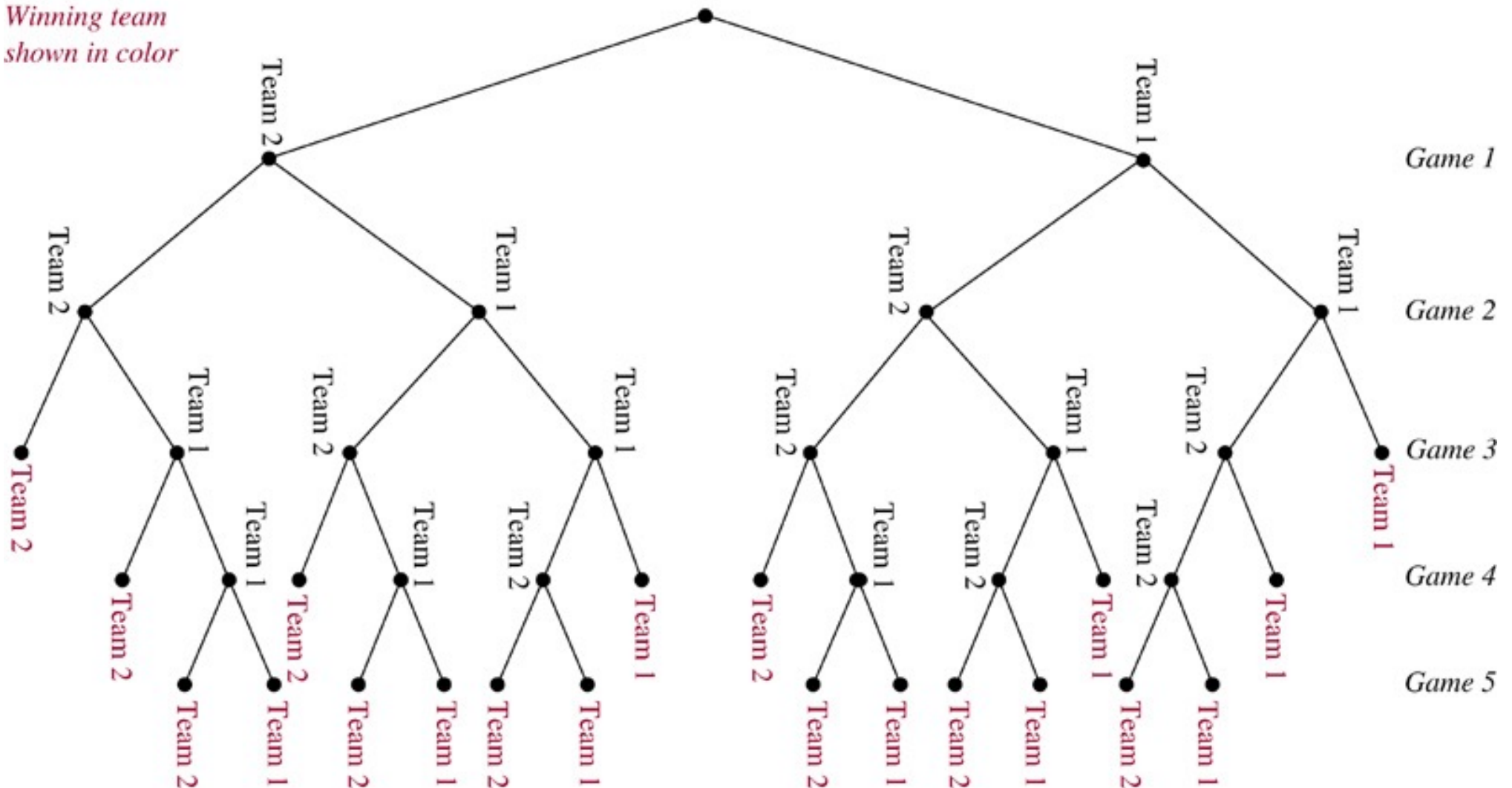


FIGURE 3 Best Three Games Out of Five Playoffs.

Ex . How many subsets of a set with 100 elements have more than one element ?

Sol:

$$(1) \binom{100}{100} + \binom{100}{99} + \binom{100}{98} + \dots + \binom{100}{2} = \dots = 2^{100} - 101$$

Thm. 4 of §4.3

$$(2) \{a_1, a_2, \dots, a_{100}\}$$

$$\text{subset: } \{\square, \square, \dots, \square\}$$

放
不放

放
不放

放
不放

$$\therefore 2^{100} - 101$$

空集合及
只有1個元素的集合

Ex 39. A **palindrome** (迴文) is a string whose reversal is identical to the string. How many bit strings of length n are palindromes ?
(abcdcba 是迴文, abcd 不是)

Sol: If $a_1 a_2 \dots a_n$ is a palindrome, then
 $a_1 = a_n, a_2 = a_{n-1}, a_3 = a_{n-2}, \dots \Rightarrow 2^{\left\lceil \frac{n}{2} \right\rceil}$ 種string.



Example. How many two-digit numbers have distinct and nonzero digits?

A two-digit number ab can be regarded as an ordered pair (a, b) , where a is the tens digit and b is the units digit. Neither of these digits is allowed to be 0 in the problem, and the two digits are to be different. There are nine choices for a , namely $1, 2, \dots, 9$. Once a is chosen, there are eight choices for b . If $a = 1$, these eight choices are $2, 3, \dots, 9$, if $a = 2$, the eight choices are $1, 3, \dots, 9$, and so on. What is important for application of the multiplication principle is that the number of choices is always 8. The answer to the questions is, by the multiplication principle, $9 \times 8 = 72$.

We can arrive at the answer 72 in another way. There are 90 two-digit numbers, $10, 11, 12, \dots, 99$. Of these numbers, nine have a 0, (namely, $10, 20, \dots, 90$) and nine have identical digits (namely, $11, 22, \dots, 99$). Thus the number of two-digit numbers with distinct and nonzero digits equals $90 - 9 - 9 = 72$. $\square$



Example. How many integers between 0 and 10,000 have only one digit equal to 5?

Let S be the set of integers between 0 and 10,000 with only one digit equal to 5.

First solution: We partition S into the set S_1 of one-digit numbers in S , the set S_2 of two-digit numbers in S , the set S_3 of three-digit numbers in S , and the set S_4 of four-digit numbers in S . There are no five-digit numbers in S . We clearly have

$$|S_1| = 1.$$

The numbers in S_2 naturally fall into two types: (1) the units digit is 5, and (2) the tens digit is 5. The number of the first type is 8 (the tens digit cannot be 0 nor can it be 5). The number of the second type is 9 (the units digit cannot be 5). Hence,

$$|S_2| = 8 + 9 = 17.$$

Reasoning in a similar way, we obtain

$$|S_3| = 8 \times 9 + 8 \times 9 + 9 \times 9 = 225, \text{ and}$$

$$|S_4| = 8 \times 9 \times 9 + 8 \times 9 \times 9 + 8 \times 9 \times 9 + 9 \times 9 \times 9 = 2673.$$

Thus,

$$|S| = 1 + 17 + 225 + 2673 = 2916.$$

Second solution: By including leading zeros (e.g., think of 6 as 0006, 25 as 0025, 352 as 0352), we can regard each number in S as a four-digit number. Now we partition S into the sets S'_1, S'_2, S'_3, S'_4 according to whether the 5 is in the first, second, third, or fourth position. Each of the four sets in the partition contains $9 \times 9 \times 9 = 729$ integers, and so the number of integers in S equals

$$4 \times 729 = 2916.$$



§6.2 The Pigeonhole Principle (鴿籠原理)

Theorem 1 (The Pigeonhole Principle)

If $k+1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

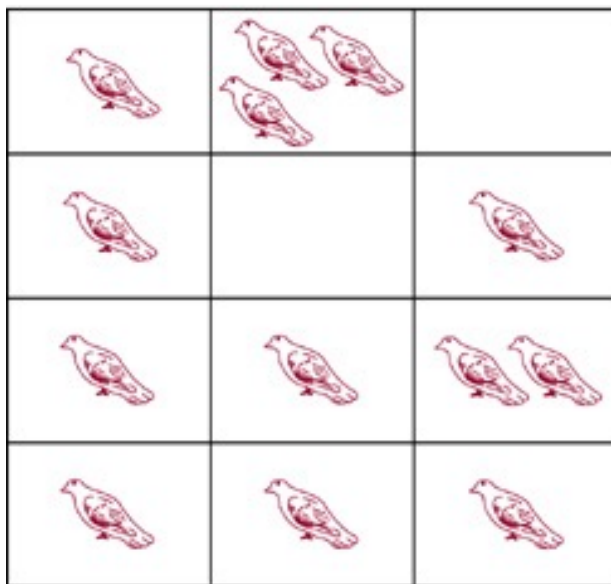
Proof

Suppose that none of the k boxes contains more than one object. Then the total number of objects would be at most k . This is a contradiction.

Example 1. Among any 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

FIGURE 1 (6 2)

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(a)



(b)



(c)

FIGURE 1 There Are More Pigeons Than Pigeonholes.



Example 2 In any group of 27 English words, there must be at least two that begin with the same letter.

Example 3 How many students must be in a class to guarantee that at least two students receive the same score on the final exam ? (0~100 points)

Sol: 102. (101+1)

Theorem 2. (The generalized pigeonhole principle)

If $\underline{N}$ objects are placed into $\underline{k}$ boxes, then there is at least one box containing at least $\left\lceil \frac{N}{k} \right\rceil$ objects.

Proof: $k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N$

e.g. 21 objects, 10 boxes $\Rightarrow$ there must be one box containing at least $\left\lceil \frac{21}{10} \right\rceil = 3$ objects.



Example 5 Among **100 people** there are at least who were **born in the same month**.

(100 objects, 12 boxes)

$$\left\lceil \frac{100}{12} \right\rceil = 9$$

Example 6 What is **minimum of students required** in a discrete mathematics class to be sure that **at least six** will receive the same grade if there are **five possible grades**, A, B, C, D, and F ?

$$5 < \left(\left\lceil \frac{N}{5} \right\rceil = 6 \right) < 7 \Rightarrow (25 + 1) \leq N < (35)$$

The Generalized Pigeonhole Principle

Example 7: a) How many cards must be selected from a standard deck of 52 cards to **guarantee that at least three cards** of the same suit are chosen?

b) How many must be selected to **guarantee that at least three hearts** are selected?

Solution:

a) We assume four boxes; one for each suit. Using the generalized pigeonhole principle, at least one box contains at least $\lceil N/4 \rceil$ cards. At least three cards of one suit are selected if $\lceil N/4 \rceil \geq 3$. The smallest integer N such that $\lceil N/4 \rceil \geq 3$ is $N = 2 \cdot 4 + 1 = 9$.

b) A deck contains 13 hearts and 39 cards which are not hearts. So, if we select 41 cards, we may have 39 cards which are not hearts along with 2 hearts. However, when we select 42 cards, we must have at least three hearts.

(Note that the generalized pigeonhole principle is not used here.)



Example 8 What is the **least number of area codes** needed to **guarantee that the 25 million phones in a state** can be assigned distinct 10-digit telephone number ? (Assume that telephone number are of the form NXX-NXX-XXXX, where the first three digits form the area code, N represents a digit from 2 to 9 inclusive, and X represents any digit.)

For each one area code, there are 8 million different phone numbers of the form “NXX-XXXX”

$$8 \cdot 10^6 = 8000000$$

$$\therefore \left\lceil \frac{25000000}{8000000} \right\rceil = 4$$

Example 10 During a month with 30 days a baseball team plays at least 1 game a day, but no more than 45 games. Show that there must be a period of some number of consecutive days during which the team must play exactly 14 games.

day	1	2	3	4	5	...	15		30
# of game	3	2	1						2

$\underbrace{\hspace{10em}}_{\text{存在一段時間的game數和}=14} \Rightarrow \text{sum} \leq 45$



Sol:

Let a_j be the number of games played on or before the j th day of the month. (第1天~第 j 天的比賽數和) ($1 \leq j \leq 30$)

Then $a_1, a_2, \dots, a_{30}$ is an increasing sequence of distinct integers with $1 \leq a_j \leq 45 \quad \forall j$

(i.e., $1 \leq a_1 < a_2 < a_3 < \dots < a_{30} \leq 45$)

Moreover, $a_1 + 14, a_2 + 14, \dots, a_{30} + 14$ is also an increasing sequence of distinct integers with $15 \leq a_j + 14 \leq 59$

(i.e., $15 \leq a_1 + 14 < a_2 + 14 < a_3 + 14 < \dots < a_{30} + 14 \leq 59$)

There are 60 positive integers $a_1, \dots, a_{30}, a_1 + 14, \dots, a_{30} + 14$ between 1 and 59. Hence, $\exists i$ and j such that

$a_i = a_j + 14$ (i.e., 第 $j+1$ 天~第 i 天 共14場)
#



§6.3 Permutations(排列) and Combinations(組合)

Def. A permutation of a set of **distinct objects** is an **ordered arrangement** of these objects. An ordered arrangement of r elements of a set is called an r -permutation.

$$P(n, r) = n \cdot (n - 1) \cdot (n - 1) \dots (n - r + 1) = \frac{n!}{(n-r)!} = x \cdot r!$$

Def. An r -combination of elements of a set is an **unordered selection** of r elements from the set.

$$C(n, c) = C_r^n = {}_n C_r = \binom{n}{r} = \frac{P(n, r)}{r!} = \frac{n!}{(n - r)! \cdot r!}$$

§6.3 Permutations(排列) and Combinations(組合)

Def. A permutation of a set of distinct objects is an ordered

1. 教練想從甲、乙、丙、丁、戊五位籃球選手中，選出三位上場參加三對三籃球賽，則選擇的方案共有多少種？
2. 教練想從甲、乙、丙、丁、戊五位籃球選手中，選出一位後衛，一位前鋒和一位中鋒，參加三對三籃球賽，則安排的方案共有多少種？



Def. An r -combination of elements of a set is an unordered selection of r elements from the set.

$$C_r^n = C(n, r) = \binom{n}{r} = \frac{p(n, r)}{r!} = \frac{n!}{r!(n-r)!}$$

§6.3 Permutations(排列) and Combinations(組合)

Def. A permutation of a set of distinct objects is an ordered arrangement of these objects. An ordered arrangement of r elements of a set is called an r -permutation.

Example 2. Let $S = \{1, 2, 3\}$.

The arrangement 3,1,2 is a permutation of S .

The arrangement 3,2 is a 2-permutation of S .

Theorem 1. The number of r -permutation of a set with n distinct elements is $P(n, r) = n \cdot (n-1) \cdot (n-2) \cdots (n-r+1) = \frac{n!}{(n-r)!}$

位置： 1 2 3 ... r
 □ □ □ ... □
 ↑ ↑ ↑ ↑
放法： n $n-1$ $n-2$... $n-r+1$



Example 4. How many different ways are there to select a first-prize winner (第一名), a second-prize winner, and a third-prize winner from 100 different people who have entered a contest ?

Sol:

$$P(100,3) = 100 \times 99 \times 98 = 970200$$

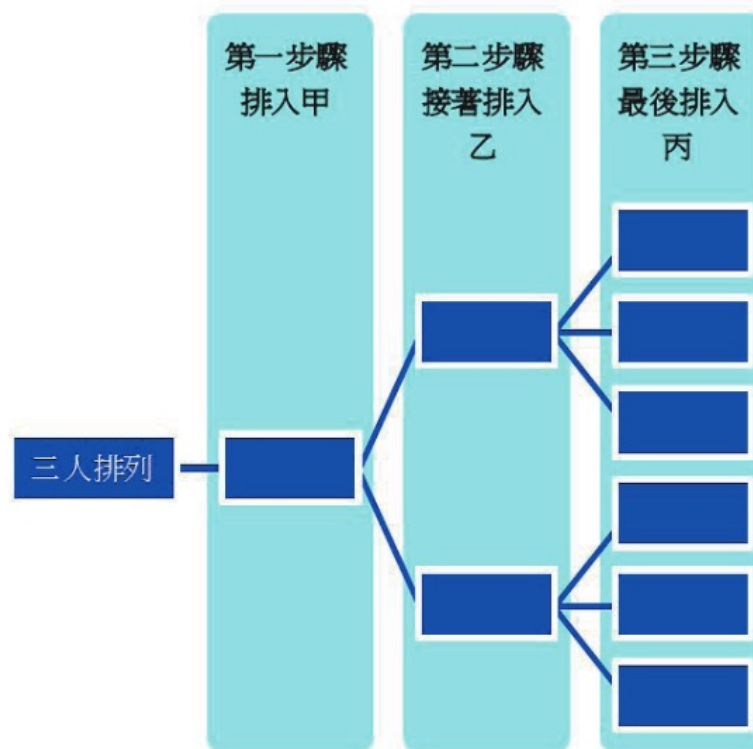
Example 6. Suppose that a saleswoman has to visit 8 different cities. She must begin her trip in a specified city, but she can visit the other cities in any order she wishes. How many possible orders can the saleswoman use when visiting these cities ?

Sol:

$$7! = 5040$$

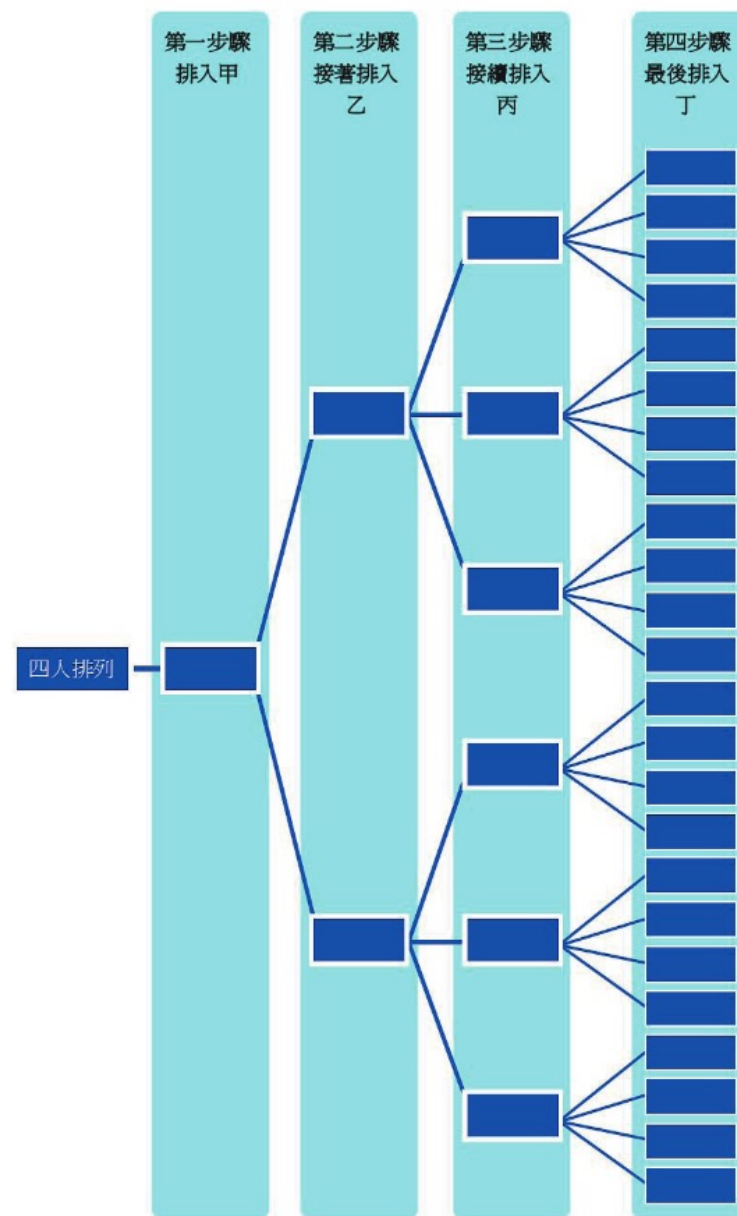
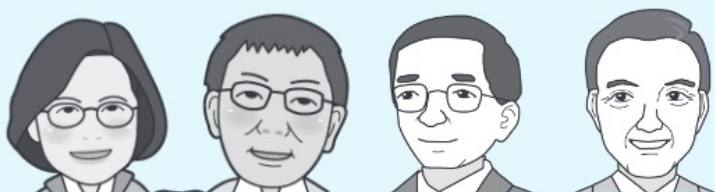
【三人排列的情形】

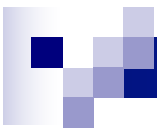
三位總統候選人甲、乙、丙於辯論會開始之前，排成一列拍照紀念，試問主辦單位共有幾種不同的排法？



【四人排列的情形】

若有四位總統候選人甲、乙、丙、丁，於辯論會開始之前，排成一列拍照紀念，試問主辦單位共有幾種不同的排法？





Solving Counting Problems by Counting Permutations

Example: How many permutations of the letters *ABCDEFGH* contain the string “*ABC*” ?

Solution: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$



Def. An r -combination of elements of a set is an unordered selection of r elements from the set.

Example 9 Let S be the set $\{1, 2, 3, 4\}$.
Then $\{1, 3, 4\}$ is a 3-combination from S .

Theorem 2 The number of r -combinations of a set with n elements, where n is a positive integer and r is an integer with $0 \leq r \leq n$, equals

$$C_r^n = C(n, r) = \binom{n}{r} = \frac{p(n, r)}{r!} = \frac{n!}{r!(n-r)!}$$

稱為 **binomial coefficient**

pf :

$$P(n, r) = C(n, r) \times r!$$



Example 10. We see that $C(4,2)=6$, since the 2-combinations of $\{a,b,c,d\}$ are the six subsets $\{a,b\}$, $\{a,c\}$, $\{a,d\}$, $\{b,c\}$, $\{b,d\}$ and $\{c,d\}$

Corollary 2. Let n and r be nonnegative integers with $r \leq n$.
Then $C(n,r) = C(n,n-r)$

pf : From Thm 2.

$$C(n,r) = \frac{n!}{r!(n-r)!} = \frac{n!}{(n-r)!(n-(n-r))!} = C(n,n-r)$$

組合意義：選 r 個拿走,相當於是選 $n - r$ 個留下.



Example 11: How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

Solution: Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$\begin{aligned} C(52, 5) &= \frac{52!}{5!47!} \\ &= \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2,598,960 \end{aligned}$$

The different ways to select 47 cards from 52 is

$$C(52, 47) = \frac{52!}{47!5!} = C(52, 5) = 2,598,960$$



Example 12: How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school.

Solution: By Theorem 2, the number of combinations is

$$C(10, 5) = \frac{10!}{5!5!} = 252.$$

Example: A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

Solution: By Theorem 2, the number of possible crews is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775$$



Example 15. Suppose there are 9 faculty members in the math department and 11 in the computer science department. How many ways are there to select a committee if the committee is to consist of 3 faculty members from the math department and 4 from the computer science department?

Sol:

$$C(9,3) \times C(11,4)$$



Ex 10. How many subsets with an odd number of elements does a set with 10 elements have ?

Sol: $C(10,1) + C(10,3) + C(10,5) + C(10,7) + C(10,9) = 512$

Ex 12. A coin is flipped 10 times where each flip comes up either heads or tails. How many possible outcomes

(a) are there in total ?

$$2^{10} = 1024$$

(b) contain exactly two heads?

$$\frac{10!}{2!8!} = 45$$

(c) contain most three tails?

$$C(10,0) + C(10,1) + C(10,2) + C(10,3) = 176$$

(d) contain the same number of heads or tails ?

$$\frac{10!}{5!5!} = 252$$



Ex 25. How many bit strings contain exactly **eight 0s** and **ten 1s** if *every 0 must be immediately followed by a 1* ?

Sol:

“1” $\times$ 2 and “01” $\times$ 8 are permuted

$$\frac{10!}{2!8!} = 45$$

§6.4 Binomial Coefficients (二項式係數)

Example 1.

$$(x+y)^3 = (x+y)(x+y)(x+y) = ?x^3 + ?x^2y + ?xy^2 + y^3$$

要產生 xy^2 項時，

需從三個括號中選兩個括號提供 y ，剩下一個則提供 x
(注意：同一個括號中的 x 跟 y 不可能相乘)

∴共有 $\binom{3}{2}$ 種不同來源的 xy^2 $\Rightarrow xy^2$ 的係數 = $\binom{3}{2}$

$$\therefore (x+y)^3 = \binom{3}{0}x^3 + \binom{3}{1}x^2y + \binom{3}{2}xy^2 + \binom{3}{3}y^3$$

Theorem 1. (The Binomial Theorem, 二項式定理)

Let x, y be variables, and let n be a positive integer, then

$$(x+y)^n = \binom{n}{0}x^n + \binom{n}{1}x^{n-1}y + \dots + \binom{n}{n-1}xy^{n-1} + \binom{n}{n}y^n = \sum_{j=0}^n \binom{n}{j}x^{n-j}y^j$$



Binomial Theorem: Let x and y be variables, and n a nonnegative integer. Then:

$$(x + y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n.$$

Proof: We use combinatorial reasoning . The terms in the expansion of $(x + y)^n$ are of the form $x^{n-j} y^j$ for $j = 0, 1, 2, \dots, n$. To form the term $x^{n-j} y^j$, it is necessary to choose $n-j$ x s from the n sums. Therefore, the coefficient of $x^{n-j} y^j$ is $\binom{n}{n-j}$ which equals $\binom{n}{j}$.

EXAMPLE 2 What is the expansion of $(x + y)^4$?



Solution: From the Binomial Theorem it follows that

$$\begin{aligned}(x + y)^4 &= \sum_{j=0}^4 \binom{4}{j} x^{4-j} y^j \\&= \binom{4}{0} x^4 + \binom{4}{1} x^3 y + \binom{4}{2} x^2 y^2 + \binom{4}{3} x y^3 + \binom{4}{4} y^4 \\&= x^4 + 4x^3 y + 6x^2 y^2 + 4x y^3 + y^4.\end{aligned}$$

EXAMPLE 3 What is the coefficient of $x^{12} y^{13}$ in the expansion of $(x + y)^{25}$?

Solution: From the Binomial Theorem it follows that this coefficient is

$$\binom{25}{13} = \frac{25!}{13! 12!} = 5,200,300.$$

Example 4. What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x-3y)^{25}$?

Sol: $(2x-3y)^{25} = (2x+(-3y))^{25}$

$$\therefore \binom{25}{13} \cdot 2^{12} \cdot (-3)^{13}$$

Cor 1. Let n be a positive integer. Then

$$\sum_{k=0}^n \binom{n}{k} = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$$

pf: By Thm 1, let $x = y = 1$

$$2^n = (1+1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{n-k} = \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}$$

Cor 2. Let n be a positive integer. Then $\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$

pf: by Thm 1. $(1-1)^n = 0$

試求 $(1.1)^8$ 的近似值至小數點後第二位。

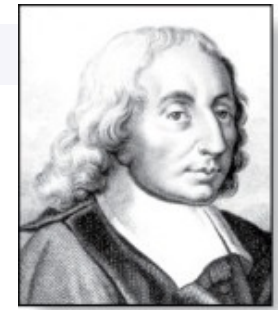
[解]:

$$\begin{aligned}(1.1)^8 &= (1 + 0.1)^8 \\&= 1 + C_1^8 \times 0.1 + C_2^8 \times 0.1^2 + C_3^8 \times 0.1^3 \\&\quad + \cdots + C_8^8 \times 0.1^8 \\&= 1 + 0.8 + 28 \times 0.01 + 56 \times 0.001 + \cdots \\&\doteq 1.8 + 0.28 + 0.056 = 2.14\end{aligned}$$

Pascal's Identity



宋 楊輝
(1261)



Blaise Pascal
(1623-1662)

Pascal's Identity: If n and k are integers with $n \geq k \geq 0$, then

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof (combinatorial): Let T be a set where $|T| = n + 1$, $a \in T$, and $S = T - \{a\}$. There are $\binom{n+1}{k}$ subsets of T containing k elements.

Each of these subsets either:

- contains a with $k - 1$ other elements, or
- contains k elements of S and not a .

There are

- $\binom{n}{k-1}$ subsets of k elements that contain a , since there are $\binom{n}{k-1}$ subsets of $k - 1$ elements of S ,
- $\binom{n}{k}$ subsets of k elements of T that do not contain a , because there are $\binom{n}{k}$ subsets of k elements of S .

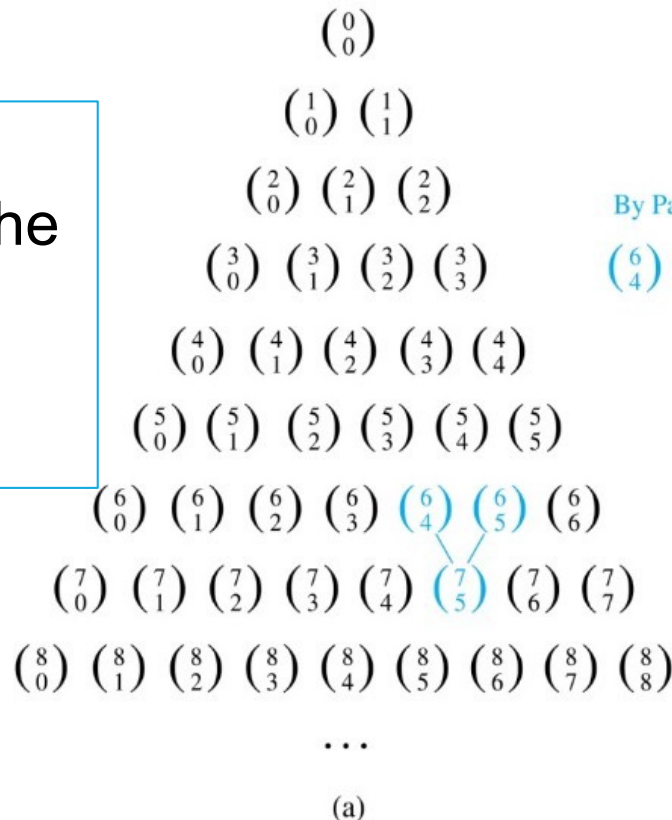
Hence,
$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

Pascal's Triangle

The n_{th} row in the triangle consists of the binomial coefficients

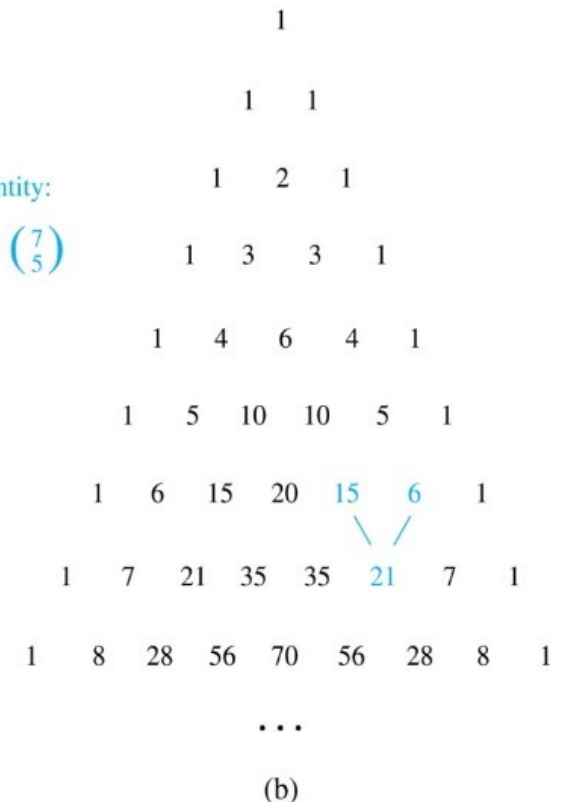
$$\binom{n}{k}, \quad k = 0, 1, \dots, n.$$

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$



By Pascal's identity:

$$\binom{6}{4} + \binom{6}{5} = \binom{7}{5}$$



By Pascal's identity, adding **two adjacent binomial coefficients** results is **the binomial coefficient in the next row** between these two coefficients.



$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$$

pf: ①(algebraic proof, 代數證明)

$$\binom{n+1}{k} = \frac{(n+1)!}{k!(n+1-k)!}$$

$$\binom{n}{k-1} + \binom{n}{k} = \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!}$$

$$= \frac{k \times n!}{k!(n-k+1)!} + \frac{(n-k+1) \times n!}{k!(n-k+1)!} = \frac{(n+1) \times n!}{k!(n-k+1)!}$$

②(combinatorial proof, 組合意義證明):

$$\begin{array}{c} n \quad 1 \\ \bigcirc \quad \cdot \\ \swarrow \searrow \\ k \end{array} \quad \text{取法} = \quad \begin{array}{c} n \quad 1 \\ \bigcirc \quad \cdot \\ \downarrow \quad \downarrow \\ k \quad 0 \end{array} + \begin{array}{c} n \quad 1 \\ \bigcirc \quad \cdot \\ \downarrow \quad \downarrow \\ k-1 \quad 1 \end{array} \Rightarrow \binom{n+1}{k} = \binom{n}{k-1} \binom{1}{0} + \binom{n}{k} \binom{1}{1}$$

Theorem 3. (Vandermode's Identity)

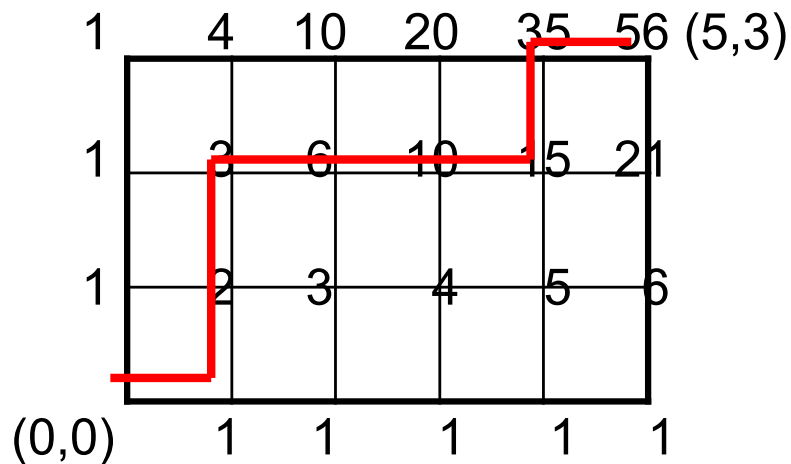
$$m, n, r \in \mathbb{Z}^+, \quad 0 \leq r \leq m, n$$

$$C(m+n, r) = \sum_{k=0}^r C(m, r-k) \cdot C(n, k)$$

pf:

$$\begin{aligned}
 C(m+n, r) &= \begin{array}{c} \text{Diagram: Two circles labeled } m \text{ and } n \text{ with arrows pointing to a single point labeled } r. \end{array} \\
 &= \begin{array}{c} \begin{array}{ccc} m & n & \\ \downarrow & \downarrow & \\ r, 0 & r-1, 1 & \dots + \downarrow \downarrow \\ & & 0, r \end{array} \end{array} \\
 &= \binom{m}{r} \binom{n}{0} + \binom{m}{r-1} \binom{n}{1} + \dots + \binom{m}{0} \binom{n}{r} \\
 &= \sum_{k=0}^r C(m, r-k) \cdot C(n, k)
 \end{aligned}$$

Ex 33. Here we will count the number of paths between the origin $(0,0)$ and point (m, n) such that each path is made up of a series of steps, where each step is a move one unit to the right or a move one unit upward.



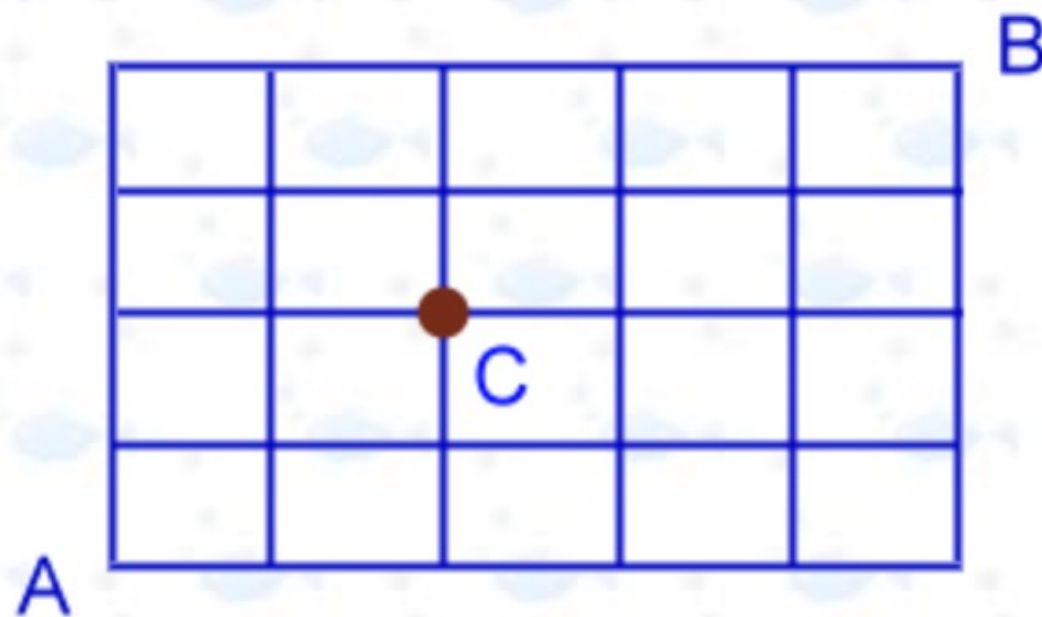
Each path can be represented by a bit string consisting of m 0s and n 1s.
 $(\rightarrow)$ $(\uparrow)$

Red path 對應的字串: 0 1 1 0 0 0 1 0

$$\binom{5+3}{5} = \frac{8!}{5!3!} = 56$$

There are $\binom{m+n}{n}$ paths of the desired type.

如下之街道圖，某人欲從A走捷徑至B但不經過C，試問共有幾種走法？



- **Ex.** Evaluate the sum

$$\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n} \quad \sum_{i=1}^n i \binom{n}{i}$$

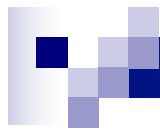
- **Ex.** Prove that

$$\binom{2n}{n} = \sum_{i=0}^n \binom{n}{i}^2 \quad \sum_{i=0}^n \binom{n}{i} \binom{n}{i} = \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i}$$

- **Ex** A farmer decides to test four fertilizer for his soybean fields. He buys **32 bags of fertilizer**, *8 bags from each kind*, and tries them randomly on **32 plots**, *8 plots from each of fields A, B, C, and D*, **one bag per plot**. If from **type I fertilizer** 1 bag is tried on field A and 3 on field B, what is the probability that two bags of type I fertilizer are tried **on field D** ?

Sol:

$$P(A) = \frac{\binom{8}{1} \cdot \binom{24}{7} \cdot \binom{7}{3} \cdot \binom{17}{5} \cdot \left[\binom{4}{2} \cdot \binom{12}{6} \right] / \binom{32}{8}}{\binom{8}{1} \cdot \binom{24}{7} \cdot \binom{7}{3} \cdot \binom{17}{5} \cdot \left[\sum_{i=0}^4 \binom{4}{i} \cdot \binom{12}{8-i} \right] / \binom{32}{8}} = \frac{\binom{4}{2} \binom{12}{6}}{\binom{16}{8}}$$



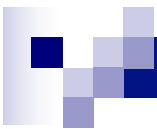
6.5 Generalized Permutations and Combinations

Permutations with Repetition

Combinations with Repetition

Permutations with Indistinguishable Objects

Distributing Objects into Boxes



Permutations with Repetition

Theorem 1: The number of r -permutations of a set of n objects with repetition allowed is n^r .

Proof: There are n ways to select an element of the set for each of the r positions in the r -permutation when repetition is allowed. Hence, by the product rule there are n^r r -permutations with repetition.

Example 1: How many strings of length r can be formed from the uppercase letters of the English alphabet?

Solution: The number of such strings is 26^r , which is the number of r -permutations of a set with 26 elements.

生活中的實例4

將5種不同的酒倒入4個不同杯子中, 若酒只能倒一次, 且每個杯子可有多種酒。試問共有幾種倒法?

[解]:將酒依序倒入杯子中, 因每種酒只能從4個不同的杯子中選一杯子倒, 均有四種可能的倒法,

$$4 \times 4 \times 4 \times 4 \times 4 = 4^5 = 1024$$

將5種不同的酒倒入4個不同杯子中, 每個杯子只能裝一種酒, 酒可以重複倒。試問共有幾種倒法?

$$5 \times 5 \times 5 \times 5 = 5^4 = 625$$



Example. How many eight-letter words can be constructed by using the 26 letters of the alphabet if each word contains three, four, or five vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

We count the number of words according to the number of vowels they contain and then use the addition principle.

First, consider words with three vowels. The three positions occupied by the vowels can be chosen in $\binom{8}{3}$ ways; the other five positions are occupied by consonants. The vowel positions can then be completed in 5^3 ways and the consonant positions in 21^5 ways. Thus, the number of words with three vowels is

$$\binom{8}{3} 5^3 21^5 = \frac{8!}{3!5!} 5^3 21^5.$$

In a similar way, we see that the number of words with four vowels is

$$\binom{8}{4} 5^4 21^4 = \frac{8!}{4!4!} 5^4 21^4,$$

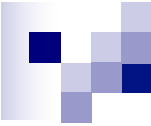
and the number of words with five vowels is

$$\binom{8}{5} 5^5 21^3 = \frac{8!}{5!3!} 5^5 21^3.$$

Hence, the total number of words is

$$\frac{8!}{3!5!} 5^3 21^5 + \frac{8!}{4!4!} 5^4 21^4 + \frac{8!}{5!3!} 5^5 21^3.$$

□



Combinations with Repetition

EXAMPLE 2 How many ways are there to select four pieces of fruit from a bowl containing apples, oranges, and pears if the order in which the pieces are selected does not matter, only the type of fruit and not the individual piece matters, and there are at least four pieces of each type of fruit in the bowl?

Solution: To solve this problem we list all the ways possible to select the fruit. There are 15 ways:

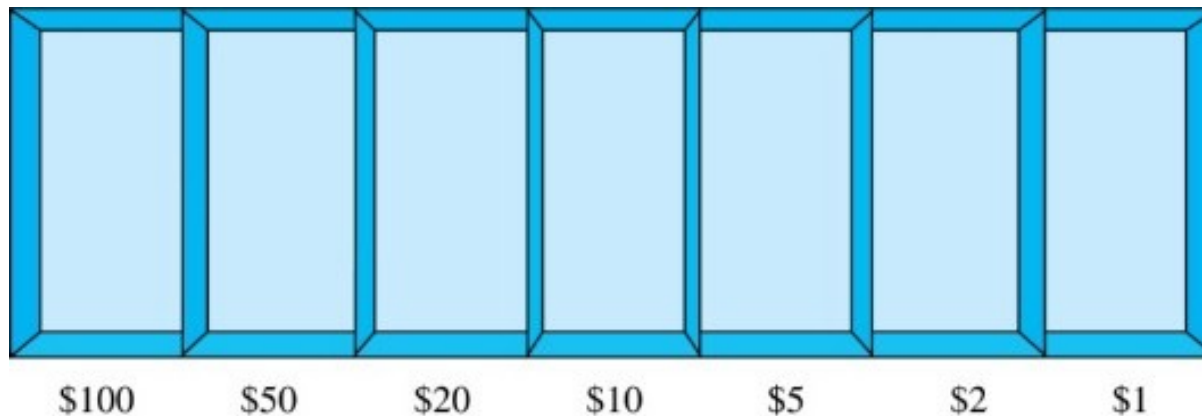
4 apples	4 oranges	4 pears
3 apples, 1 orange	3 apples, 1 pear	3 oranges, 1 apple
3 oranges, 1 pear	3 pears, 1 apple	3 pears, 1 orange
2 apples, 2 oranges	2 apples, 2 pears	2 oranges, 2 pears
2 apples, 1 orange, 1 pear	2 oranges, 1 apple, 1 pear	2 pears, 1 apple, 1 orange

The solution is the number of 4-combinations with repetition allowed from a three-element set, $\{apple, orange, pear\}$. 

Combinations with Repetition

Example 3: How many ways are there to select five bills from a box containing at least five of each of the following denominations: \$1, \$2, \$5, \$10, \$20, \$50, and \$100?

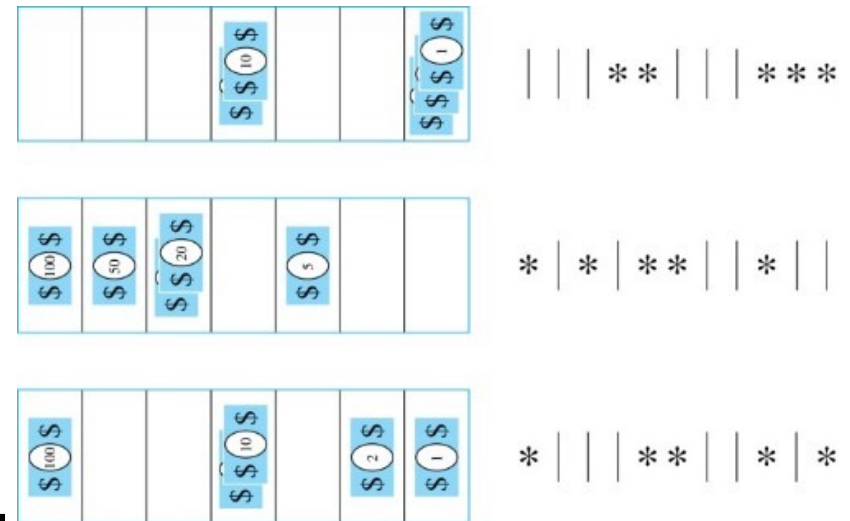
Solution: Place the selected bills in the appropriate position of a cash box illustrated below:



Combinations with Repetition

Some possible ways of placing the five bills:

The number of ways to select five bills corresponds to the number of ways to arrange six bars and five stars in a row.



This is the number of unordered selections of 5 objects from a set of 11. Hence, there are

$$C(11, 5) = \frac{11!}{5!6!} = 462$$

ways to choose five bills with seven types of bills.



Combinations with Repetition

Theorem 2: The number of r -combinations from a set with n elements when repetition of elements is allowed is

$$\binom{n + r - 1}{n} = \binom{n + r - 1}{r - 1}$$

Proof: Each r -combination of a set with n elements with repetition allowed can be represented by a list of $r - 1$ bars and n stars. The bars mark the r cells containing a star for each time the i th element of the set occurs in the combination.

The number of such lists is $C(n + r - 1, n)$, because each list is a choice of the r positions to place the stars, from the total of $n + r - 1$ positions to place the stars and the bars. This is also equal to $C(n + r - 1, r - 1)$, which is the number of ways to place the $r - 1$ bars.

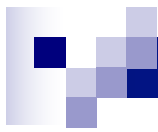
Thm 2. There are $C(n+r-1, n)$ r -combinations from a set with n elements when repetition of elements is allowed.

pf: (視為有 n 個 $*$ ，要放入 r 個區域，
故需插入 $r-1$ 個 **bar** 將這些 $*$ 隔成 r 區)

例：設 $r = 4$, 集合為 $\{a_1, a_2, a_3, a_4\}$, $n = 6$

a_2 出現 1 次 a_3 不出現
 $\underbrace{**}_{a_1 \text{ 出現 2 次}} | * | | \underbrace{***}_{a_4 \text{ 出現 3 次}}$ $\xRightarrow{\text{取法}}$ $\{a_1, a_1, a_2, a_4, a_4, a_4\}$

$\therefore \binom{n+r-1}{n}$ 種方法_#



Example 4: Suppose that a cookie shop has **four different kinds of cookies**. How many different ways can **six cookies be chosen**?

Solution: The number of ways to choose six cookies is the number of 6-combinations of a set with four elements. By Theorem 2

$$C(9, 6) = C(9, 3) = \frac{9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3} = 84$$

is the number of ways to choose six cookies from the four kinds.



Example 5: How many solutions does the equation

$$x_1 + x_2 + x_3 = 11$$

have, where x_1 , x_2 and x_3 are nonnegative integers?

Solution: Each solution corresponds to a way to **select 11 items** from a set with **three elements**; x_1 elements of type one, x_2 of type two, and x_3 of type three.

By Theorem 2 it follows that there are

$$C(3+11-1, 11) = C(13, 11) = C(13, 2) = \frac{13 \cdot 2}{1 \cdot 2} = 78$$

solutions.

※若上題改要求 $x_1 \geq 1, x_2 \geq 2, x_3 \geq 3$,

則原式 $x_1 + x_2 + x_3 = 11$

可改成 $(x_1 - 1) + (x_2 - 2) + (x_3 - 3) = 11 - 1 - 2 - 3 = 5$

此時相當於是 $y_1 + y_2 + y_3 = 5$

其中 $y_1 = x_1 - 1, y_2 = x_2 - 2, y_3 = x_3 - 3$ 且 $y_1, y_2, y_3 \in \mathbb{N}$

∴ 5個1間要插入2個bar $\Rightarrow \binom{5+2}{5}$ 種

(注意： **case** $y_1 = 1, y_2 = 3, y_3 = 1$ 相當於 $x_1 = 2, x_2 = 5, x_3 = 4$)

※若上題再改為 $1 \leq x_1 \leq 3, x_2 \geq 2, x_3 \geq 3$ ，則需排除 $x_1 > 3$ 的情形
(即 $x_1 \geq 4$ 的情形)

因 $(x_1 - 4) + (x_2 - 2) + (x_3 - 3) = 11 - 9 = 2$

∴ 共有 $\binom{5+2}{5} - \binom{2+2}{2}$ 種



ex. How many solutions are there to the inequality

$$x_1 + x_2 + x_3 \leq 11$$

where x_1, x_2, x_3 are nonnegative integers ?

■ Sol:

[Hint: Introduce an auxiliary variable x_4 ,

such that $x_1 + x_2 + x_3 + x_4 = 11$]

$$\binom{11+4-1}{11} = \binom{14}{11}$$

8. How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 21$$

where $x_i, i = 1, 2, 3, 4, 5$, is a nonnegative integers such that

- a) $x_1 \geq 1$
- b) $x_i \geq 2$
- c) $0 \leq x_1 \leq 10$
- d) $0 \leq x_1 \leq 3, 1 \leq x_2 < 4, \text{ and } x_3 \geq 15$

■ **Sol:**

a)

$$x_1 + x_2 + x_3 + x_4 + x_5 = 20$$

$$\binom{20+5-1}{20} = \binom{24}{20} = 10626$$

8. How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 21$$

where $x_i, i = 1, 2, 3, 4, 5$, is a nonnegative integers such that

- a) $x_1 \geq 1$
- b) $x_i \geq 2$
- c) $0 \leq x_1 \leq 10$
- d) $0 \leq x_1 \leq 3, 1 \leq x_2 < 4$, and $x_3 \geq 15$

■ **Sol:**

b)

$$x_1 + x_2 + x_3 + x_4 + x_5 = 11$$

$$\binom{11+5-1}{11} = \binom{15}{11} = 1365$$

8. How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 21$$

where $x_i, i = 1, 2, 3, 4, 5$, is a nonnegative integers such that

- a) $x_1 \geq 1$
- b) $x_i \geq 2$
- c) $0 \leq x_1 \leq 10$
- d) $0 \leq x_1 \leq 3, 1 \leq x_2 < 4$, and $x_3 \geq 15$

■ Sol:

c)

$$x_1 + x_2 + x_3 + x_4 + x_5 = 21 \qquad \binom{21+5-1}{21} = \binom{25}{21} = 12650$$

$\therefore x_1 \geq 11$ has to be excluded

$$x_1 + x_2 + x_3 + x_4 + x_5 = 10 \qquad \binom{10+5-1}{10} = \binom{14}{10} = 1001$$

$$\therefore 12650 - 1001 = 11649$$

8. How many solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 + x_5 = 21$$

where $x_i, i = 1, 2, 3, 4, 5$, is a nonnegative integers

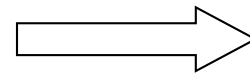
such that

- a) $x_1 \geq 1$
- b) $x_i \geq 2$
- c) $0 \leq x_1 \leq 10$
- d) $0 \leq x_1 \leq 3, 1 \leq x_2 < 4, \text{ and } x_3 \geq 15$

■ **Sol:**

d) $\because x_3 \geq 15$

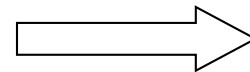
(i) $x_1 + x_2 + x_3 + x_4 + x_5 = 6 = 21 - 15$
 $\because x_3 \geq 15 \text{ \& } 1 \leq x_2 \leq 3$
 $\therefore x_1 + x_2 + x_3 + x_4 + x_5 = 5 = 21 - 15 - 1$
& then $0 \leq x_2 - 1 \leq 2$



if $x_2 \geq 1 \text{ \& } x_3 \geq 15$
all possible solution

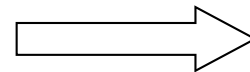
$$\binom{5+5-1}{5} = \binom{9}{5} = 126$$

(ii) $\because 0 \leq x_1 \leq 3$
 $\therefore x_1 \geq 4$ has to be excluded
 $x_1 + x_2 + x_3 + x_4 + x_5 = 5 - 4 = 1$



$x_1 \geq 4$ possible solutions
 $\binom{1+5-1}{1} = 5$

(iii) $\because 0 \leq x_2 - 1 \leq 2$
 $\therefore x_2 \geq 3$ has to be excluded
 $x_1 + x_2 + x_3 + x_4 + x_5 = 5 - 3 = 2$



$x_2 \geq 3$ possible solutions
 $\binom{2+5-1}{2} = 15$

(iv) $126 - 5 - 15 = 106$



Summarizing the Formulas for Counting Permutations and Combinations with and without Repetition

TABLE 1 Combinations and Permutations With and Without Repetition.

<i>Type</i>	<i>Repetition Allowed?</i>	<i>Formula</i>
r -permutations	No	$\frac{n!}{(n-r)!}$
r -combinations	No	$\frac{n!}{r!(n-r)!}$
r -permutations	Yes	n^r
r -combinations	Yes	$\frac{(n+r-1)!}{r!(n-1)!}$



Example. Twenty-five points are chosen in the plane so that no three of them are collinear. How many straight lines do they determine? How many triangles do they determine?

Since no three of the points lie on a line, every pair of points determines a unique straight line. Thus, the number of straight lines determined equals the number of 2-subsets of a 25-element set, and this is given by

$$\binom{25}{2} = \frac{25!}{2!23!} = 300.$$

Similarly, every three points determines a unique triangle, so that the number of triangles determined is given by

$$\binom{25}{3} = \frac{25!}{3!22!}.$$

□



Example. There are 15 people enrolled in a mathematics course, but exactly 12 attend on any given day. The number of different ways that 12 students can be chosen is

$$\binom{15}{12} = \frac{15!}{12!3!}.$$

If there are 25 seats in the classroom, the 12 students could seat themselves in $P(25, 12) = 25!/13!$ ways. Thus, there are

$$\binom{15}{12} P(25, 12) = \frac{15!25!}{12!3!13!}$$

ways in which an instructor might see the 12 students in the classroom.

□



Permutations with Indistinguishable Objects

Example: How many different strings can be made by reordering the letters of the word *SUCCESS*.

Solution: There are seven possible positions for the three *S*s, two *C*s, one *U*, and one *E*.

- The three *S*s can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two *C*s can be placed in $C(4,2)$ different ways, leaving two positions free.
- The *U* can be placed in $C(2,1)$ different ways, leaving one position free.
- The *E* can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3) \cdot C(4,2) \cdot C(2,1) \cdot C(1,1) = \frac{7!}{3! \cdot 4!} \cdot \frac{4!}{2! \cdot 2!} \cdot \frac{2!}{1! \cdot 1!} \cdot \frac{1!}{1! \cdot 0!} = 420$$

Permutations with Indistinguishable Objects

Theorem 3: The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, ..., and n_k indistinguishable objects of type k , is:

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

Proof: By the product rule the total number of permutations is:

$C(n, n_1) C(n - n_1, n_2) \cdots C(n - n_1 - n_2 - \cdots - n_{k-1}, n_k)$ since:

- The n_1 objects of type one can be placed in the n positions in $C(n, n_1)$ ways, leaving $n - n_1$ positions.
- Then the n_2 objects of type two can be placed in the $n - n_1$ positions in $C(n - n_1, n_2)$ ways, leaving $n - n_1 - n_2$ positions.
- Continue in this fashion, until n_k objects of type k are placed in $C(n - n_1 - n_2 - \cdots - n_{k-1}, n_k)$ ways.

The product can be manipulated into the desired result as follows:

$$\frac{n!}{n_1!(n - n_1)!} \frac{(n - n_1)!}{n_2!(n - n_1 - n_2)!} \cdots \frac{(n - n_1 - \cdots - n_{k-1})!}{n_k!0!} = \frac{n!}{n_1!n_2!\cdots n_k!}.$$

Example. The so-called “15 puzzle” consists of 15 sliding unit squares labeled with the numbers 1 through 15 and mounted in a 4-by-4 square frame as shown in Figure 2.1. The challenge of the puzzle is to move from the initial position shown to any specified position. (That challenge is not the subject of this problem.) By a position, we mean an arrangement of the 15 numbered squares in the frame with one empty unit square. What is the number of positions in the puzzle (ignoring whether it is possible to move to the position from the initial one)?

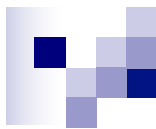
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	



Figure 2.1

The problem is equivalent to determining the number of ways to assign the numbers $1, 2, \dots, 15$ to the 16 squares of a 4-by-4 grid, leaving one square empty. Since we can assign the number 16 to the empty square, the problem is also equivalent to determining the number of assignments of the numbers $1, 2, \dots, 16$ to the 16 squares, and this is $P(16, 16) = 16!$.

What is the number of ways to assign the numbers $1, 2, \dots, 15$ to the squares of a 6-by-6 grid, leaving 21 squares empty? These assignments correspond to the 15-permutations of the 36 squares as follows: To an assignment of the numbers $1, 2, \dots, 15$ to 15 of the squares, we associate the 15-permutation of the 36 squares obtained by putting the square labeled 1 first, the square labeled 2 second, and so on. Hence the total number of assignments is $P(36, 15) = 36!/21!$. $\square$



Example. What is the number of ways to order the 26 letters of the alphabet so that no two of the vowels a, e, i, o , and u occur consecutively?

The solution to this problem (like so many counting problems) is straightforward once we see how to do it. We think of two main tasks to be accomplished. The first task is to decide how to order the consonants among themselves. There are 21 consonants, and so $21!$ permutations of the consonants. Since we cannot have two consecutive vowels in our final arrangement, the vowels must be in 5 of the 22 spaces before, between, and after the consonants. Our second task is to put the vowels in these places. There are 22 places for the a , then 21 for the e , 20 for the i , 19 for the o , and 18 for the u . That is, the second task can be accomplished in

$$P(22, 5) = \frac{22!}{17!}$$

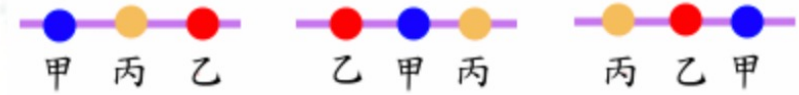
ways. By the multiplication principle, we determine that the number of ordered arrangements of the letters of the alphabet with no two vowels consecutive is

$$21! \times \frac{22!}{17!}.$$

□



(圖1)



(圖2)



直線排列

環狀排列

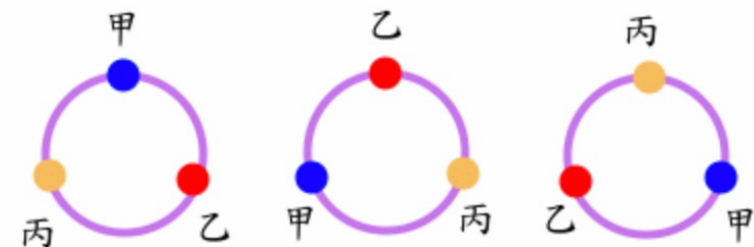
環狀排列

(1) n 個不同物件的環狀排列數為 $\frac{n!}{n} = (n-1)!$

(2) n 個不同物件中, 任取 m 個的環狀排列數為 $\frac{P_m^n}{m}$ 。



(圖3)



生活中的實例1

五對夫婦圍圓桌而坐, 試問男女相間坐的方法數為何?

[解]:

先直線排列: $5! \times 5! \times 2$

故環狀排列數為 $\frac{5! \times 5! \times 2}{10} = 4! \times 5! = 2880$ 。

生活中的實例2

五對夫婦圍圓桌而坐, 試問每對夫妻相鄰而坐的方法數為何?

[解]:

先直線排列: $5! \times 2^5$

故環狀排列數為 $\frac{5! \times 2^5}{5} = 4! \times 2^5$ 。



Example. Ten people, including two who do not wish to sit next to one another, are to be seated at a round table. How many circular seating arrangements are there?

We solve this problem using the subtraction principle. Let the 10 people be $P_1, P_2, P_3, \dots, P_{10}$, where P_1 and P_2 are the two who do not wish to sit together. Consider seating arrangements for 9 people $X, P_3, \dots, P_{10}$ at a round table. There are $8!$ such arrangements. If we replace X by either P_1, P_2 or by P_2, P_1 in each of these arrangements, we obtain a seating arrangement for the 10 people in which P_1 and P_2 are next to one another. Hence using the subtraction principle, we see that the number of arrangements in which P_1 and P_2 are not together is $9! - 2 \times 8! = 7 \times 8!$.

Another way to analyze this problem is the following: First seat P_1 at the “head” of the table. Then P_2 cannot be on either side of P_1 . There are 8 choices for the person on P_1 ’s left, 7 choices for the person on P_1 ’s right, and the remaining seats can be filled in $7!$ ways. Thus, the number of seating arrangements in which P_1 and P_2 are not together is

$$8 \times 7 \times 7! = 7 \times 8!.$$

生活中的實例3

有8個不同顏色的珠子, 全部串成一項圈, 試問其方法數有多少種?

[解]:

先將其想像成8個不同的物品做環狀排列, 故其方法數有 $8!/8 = 7! = 5040$,

其次因項鍊沒有正反面之分, 故須除以2, 所以共有 $5040/2 = 2520$ 種。

隨堂練習3

有8個不同顏色的珠子, 取6個串成一項鍊, 試問其方法數有多少種?

[解]:1680種。

$$\frac{P_6^8}{6} \times \frac{1}{2} = \frac{8!}{2!} \times \frac{1}{6} \times \frac{1}{2} = 1680$$



Example. What is the number of necklaces that can be made from 20 beads, each of a different color?

There are $20!$ permutations of the 20 beads. Since each necklace can be rotated without changing the arrangement of the beads, the number of necklaces is at most $20!/20 = 19!$. Since a necklace can also be turned over without changing the arrangement of the beads, the total number of necklaces, by the division principle, is $19!/2$.

□



Distributing Objects into Boxes

Many counting problems can be solved by counting the ways objects can be placed in boxes.

- The objects may be either different from each other (*distinguishable*) or identical (*indistinguishable*).
- The boxes may be labeled (*distinguishable*) or unlabeled (*indistinguishable*).

Distributing objects into Boxes

Distinguishable objects and distinguishable boxes

Example 8. How many ways are there to distribute hands of 5 cards to each of four players from the standard deck of 52 cards?

Sol: player 1 : $\binom{52}{5}$ 種
player 2 : 從剩下的牌再發5張 $\Rightarrow \binom{47}{5}$

$$\therefore \binom{52}{5} \binom{47}{5} \binom{42}{5} \binom{37}{5} = \frac{52!}{5!47!} \cdot \frac{47!}{5!42!} \cdot \frac{42!}{5!37!} \cdot \frac{37!}{5!32!} = \frac{52!}{5!5!5!5!32!}$$

注意：上題相當於將52張牌放進5個不同的box的放法，
即 box 1 的給 player 1
box 2 的給 player 2
box 3 的給 player 3
box 4 的給 player 4
而 box 5 的是剩下的牌。

Thm 4. The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i , $i=1, 2, \dots, k$, equals

$$\frac{n!}{n_1!n_2!\cdots n_k!} \quad (\text{跟Thm 3相等})$$

Indistinguishable objects and distinguishable boxes

Example 9. How many ways are there to place 10 indistinguishable balls into eight distinguishable bins?

Sol: $C(10+8-1, 10)$

$\Rightarrow$ There are $C(n+r-1, r-1)$ ways to place n indistinguishable objects into r distinguishable boxes.

Distinguishable objects and indistinguishable boxes

Example 10. How many ways are there to put four different employees into three indistinguishable offices, when each office can contain any number of employees?

Sol: employees: A, B, C, D

4人在同一office: $\{\{A, B, C, D\}\} \Rightarrow 1$ 種

3人在同一office, 1人在另一office: $\{\{A, B, C\}, \{D\}\}, \{\{A, B, D\}, \{C\}\}, \dots \Rightarrow 4$ 種

2人在同一office, 2人在另一office: $\{\{A, B\}, \{C, D\}\}, \{\{A, D\}, \{B, C\}\}, \dots \Rightarrow 3$ 種

2人在同一office, 另2人分別在剩下兩個office:

$\{\{A, B\}, \{C\}, \{D\}\}, \{\{A, D\}, \{B\}, \{C\}\}, \dots \Rightarrow 6$ 種

$\therefore$ 共 14種

Note. There is no simple closed formula.

可參考 Stirling numbers of the second kind. (p. 378)

Indistinguishable objects and indistinguishable boxes

Example 11. How many ways are there to pack six copies of the same book into four identical boxes, where a box can obtain as many as six books?

Sol:

6	3, 3	2, 2, 1, 1
5, 1	3, 2, 1	∴ 共 9種
4, 2	3, 1, 1, 1	
4, 1, 1	2, 2, 2	

Note. This problem is the same as writing n as the sum of at most k positive integers in nonincreasing order.

That is, $a_1 + a_2 + \dots + a_j = n$, where $a_1, a_2, \dots, a_j$ are positive integers with $a_1 \geq a_2 \geq \dots \geq a_j$ and $j \leq k$.

No simple closed formula exists.



Ex 16. How many ways are there to distribute 12 indistinguishable balls into six distinguishable bins ?

Sol:

$$\binom{12 + 6 - 1}{12}$$

Ex 18. How many ways are there to distribute 15 distinguishable objects into five distinguishable boxes so that the boxes have one, two, three, four, and five objects in them, respectively.

Sol:

$$\frac{15! \cdot 5!}{1! \cdot 2! \cdot 3! \cdot 4! \cdot 5!}$$



Example. How many possibilities are there for eight nonattacking rooks on an 8-by-8 chessboard?

An example of eight nonattacking rooks on an 8-by-8 board is the following:

					⊗		
							⊗
⊗							
		⊗					
				⊗			
						⊗	
	⊗						
			⊗				

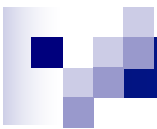
We give each square on the board a pair (i, j) of coordinates. The integer i designates the row number of the square, and the integer j designates the column number of the square. Thus, i and j are integers between 1 and 8. Since the board is 8-by-8 and there are to be eight rooks on the board that cannot attack one another, there must be exactly one rook in each row. Thus, the rooks must occupy eight squares with coordinates

$$(1, j_1), (2, j_2), \dots, (8, j_8).$$

But there must also be exactly one rook in each column so that no two of the numbers $j_1, j_2, \dots, j_8$ can be equal. More precisely,

$$j_1, j_2, \dots, j_8$$

must be a permutation of $\{1, 2, \dots, 8\}$. Conversely, if $j_1, j_2, \dots, j_8$ is a permutation of $\{1, 2, \dots, 8\}$, then putting rooks in the squares with coordinates $(1, j_1), (2, j_2), \dots, (8, j_8)$ we arrive at eight nonattacking rooks on the board. Thus, we have a one-to-one correspondence between sets of 8 nonattacking rooks on the 8-by-8 board and permutations of $\{1, 2, \dots, 8\}$. Since there are $8!$ permutations of $\{1, 2, \dots, 8\}$, there are $8!$ ways to place eight rooks on an 8-by-8 board so that they are nonattacking.



We implicitly assumed in the preceding argument that the rooks were *indistinguishable* from one another, that is, they form a multiset of eight objects all of one type. Therefore, the only thing that mattered was which squares were occupied by rooks. If we have eight distinct rooks, say eight rooks each colored with one of eight different colors, then we have also to take into account which rook is in each of the eight occupied squares. Let us thus suppose that we have eight rooks of eight different colors. Having decided which eight squares are to be occupied by the rooks ($8!$ possibilities), we now have also to decide what the color is of the rook in each of the occupied squares. As we look at the rooks from row 1 to row 8, we see a permutation of the eight colors. Hence, having decided which eight squares are to be occupied ($8!$ possibilities), we then have to decide which permutation of the eight colors ($8!$ permutations) we shall assign. Thus, the number of ways to have eight nonattacking rooks of eight different colors on an 8-by-8 board equals

$$8!8! = (8!)^2.$$

Now suppose that, instead of rooks of eight different colors, we have one red (R) rook, three blue (B) rooks, and four (Y) yellow rooks. It is assumed that rooks of the same color are indistinguishable from one another.⁸ Now, as we look at the rooks from row 1 to row 8, we see a permutation of the colors of the multiset

$$\{1 \cdot R, 3 \cdot B, 4 \cdot Y\}.$$

The number of permutations of this multiset equals, by Theorem 2.4.2,

$$\frac{8!}{1!3!4!}.$$